

Price-Insensitive Flows and Yield Curve Deformation

Causal Evidence from India's JP Morgan GBI-EM Index Inclusion

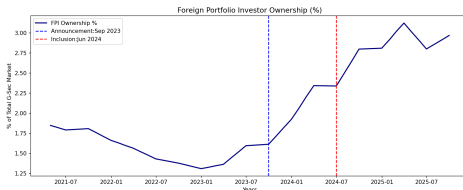
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Motivation and Research Question:

- ▶ On **21 September 2023**, JP Morgan announced that Indian government bonds under the **Fully Accessible Route (FAR)** would join the GBI-EM Global Diversified index.
- ▶ India's weight rose mechanically by **one percentage point a month**, from June 2024 to the **10% cap** in March 2025.
- ▶ Passive funds must hold Indian bonds in proportion to India's weight: as it climbs, they buy. This demand is **insensitive to price**.
- ▶ **Research question:** did index-driven foreign buying change the **level and the shape** of India's sovereign yield curve?
- ▶ **Why not a simple regression?** FPI and yields are **determined together**: foreign investors buy when yields look attractive, and global risk shocks move both at once.



FPI ownership of Indian government securities: 1.3% in early 2023, 2.3% at the start of inclusion, 3.1% in March 2025.

Hypotheses, Data and Institutional Setting:

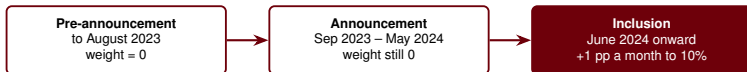
Hypotheses

- ▶ **H1 Level:** index-driven FPI lowers sovereign yields.
- ▶ **H2 Heterogeneity:** the effect differs across maturities, not a parallel shift.
- ▶ **H3 Eligibility:** compression is concentrated in index-eligible maturities (≥ 2.5 years).
- ▶ **H4 Regime:** inclusion changes the probability of a flat-curve regime.
- ▶ **H5 Instrument:** the pre-announced weight schedule is a valid instrument.

Data

- ▶ Monthly, April 2021 – January 2026; baseline regression May 2021 – March 2025, 47 observations.
- ▶ G-Sec yields at 13 maturities, 1 to 30 years (RBI); FPI ownership (RBI); JP Morgan index weight.
- ▶ **Global controls:** US 10Y and 2Y yields, VIX, dollar index, Brent.
- ▶ **Domestic controls:** repo rate, CPI (level), net LAF liquidity.

Timeline



Empirical Strategy — Instrumental Variables:

The endogeneity problem

- ▶ OLS is biased: inflows raise bond prices, but **high yields also attract inflows**; global risk appetite moves flows and yields together. The first channel biases OLS **toward zero**.



First stage

$$FPI_t = \pi_0 + \pi_1 W_t + X_t' \phi + \nu_t$$

- ▶ +1 pp of weight $\Rightarrow$ FPI **+0.084 pp** (HAC $p = 0.002$).
- ▶ First-stage $F = 23.69$; $R^2 = 0.833$.

Second stage

$$\Delta y_{i,t} = \alpha + \lambda \widehat{FPI}_t + X_t' \gamma + u_t$$

- ▶ Schedule fixed in September 2023: its timing **cannot respond** to India's month-to-month conditions.
- ▶ **HAC (Newey–West)** errors, Bartlett kernel, 4 lags.

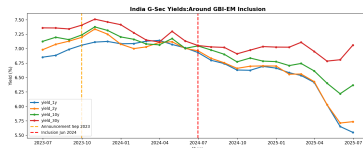
Main Result — 10-Year Yield Compression:

A 1 pp increase in FPI ownership lowers the monthly change in the 10-year yield by **14.5 basis points**.

Estimator	Coef.	p	95% CI
OLS, HAC	-0.1233	0.001	
2SLS, robust	-0.1446	0.092	[-0.313, 0.023]
2SLS, HAC	-0.1446	0.010	[-0.255, -0.034]
Anderson-Rubin		0.051	

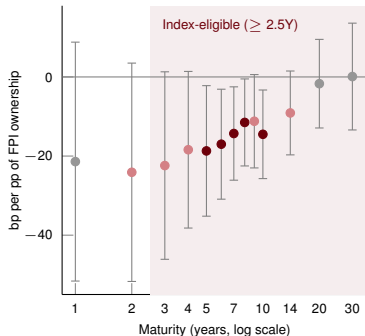
Outcome: Δ 10Y; full controls; $n = 47$.

- ▶ 2SLS is **17% larger** than OLS (-12.3 bp): consistent with OLS biased toward zero as foreign money **chases yields**.
- ▶ FPI rose **2.34% $\rightarrow$ 3.12%** over the ramp: implied compression $\approx$ **63 bp**; the 10Y actually fell **39 bp**.
- ▶ 1 pp of ownership $\approx$ **Rs 1.13 lakh crore** ($\approx$ \$13.5bn) $\Rightarrow$ about **5 bp per \$1bn** of index-driven holdings.



G-Sec yields around inclusion: steady fall through the ramp.

Yield-Curve Deformation:



2SLS effect by maturity; bars are 95% HAC intervals. Dark: $p < 0.05$;
light: $p < 0.10$; grey: not significant.

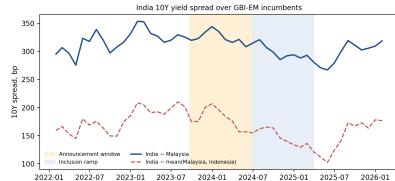
- Significant across **5–10 years**; weaker at 2–4; **zero** at 20–30 (30Y: +0.001, $p = 0.993$).
- The index **deforms** the curve: the belly falls, the ultra-long end stays **anchored**.

Spread	2SLS	HAC p
30Y – 10Y	+0.145	0.016
30Y – 2Y	+0.241	0.042
10Y – 2Y	+0.096	0.329

- **Mechanism:** index money buys liquid FAR benchmarks (**5–14Y**); 20–30Y bonds are held by **insurers and pension funds** matching long liabilities.
- **H3 rejected:** short maturities move as much (–0.185 vs –0.134 at eligible tenors), consistent with **arbitrage and liquidity**.

Robustness and Outside Evidence:

- **Specifications:** 10Y estimate stays between -0.135 and -0.180 when controls are dropped or the sample shortened.
- **Official CPI:** MoSPI CPI reproduces the baseline (-0.141); extending to December 2025 strengthens it to -0.208 (AR $p = 0.050$).
- **Timing placebo:** the same ten-month ramp placed before any index news gives **none of 13** significant (smallest $p = 0.31$).
- **Peers:** India's 10Y spread over Malaysia and Indonesia narrowed **36 bp** during the ramp and widened **56 bp** after it.
- **Regime:** each pp of weight lowers the probability of a flat curve by **8 pp** (association).



India's 10Y spread over GBI-EM incumbents; ramp shaded.

Rebalancing days ($t - 5$ to $t + 5$)

	Rebal. (bp)	Down	Other (bp)
India 10Y	-4.30	8 of 10	+3.35
India 2Y	-6.47	9 of 10	+7.40
Malaysia 10Y	-0.41	7 of 10	-0.10
Indonesia 10Y	-1.76	5 of 10	-0.50

Implications, Limitations and Conclusion:

Key findings

- ▶ 10Y: **-14.5 bp** per 1 pp of FPI ownership.
- ▶ Compression between **2 and 10 years**, absent at 20–30.
- ▶ **30Y–10Y spread widens** as the belly compresses.
- ▶ Peers and rebalancing dates **support** the IV results.

Implications

- ▶ **Investors:** the **5–10Y segment**, not the long bond, captures the index effect.
- ▶ **RBI:** the 10Y fell 39 bp with repo at 6.50%: conditions **eased ahead of policy**.
- ▶ **Debt management:** **5–14Y issuance** benefits most.

Limitations

- ▶ Only **47 months**; the instrument varies in only **ten**.
- ▶ HAC first-stage $F = 9.3$ is below 10, though Anderson–Rubin still rejects at 5.1%.
- ▶ Estimate **depends on the CPI-level** control.
- ▶ **Front-running** in the announcement window is not captured by the instrument.

India's inclusion in the JP Morgan GBI-EM compressed long-term sovereign yields through **price-insensitive foreign demand**. The effect is a **deformation, not a parallel shift**.

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